

XD Inc. : Bringing quality games to players worldwide with highly stable network connectivity

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About XD Inc.

XD uses the external high-performance network of Google Cloud to establish the best connectivity for games around the world, complying with local regulations.

XD Inc. is a Chinese game developer and distributor that has been offering web and mobile games since 2009. Currently operating 38 free and paid games, XD has 50 million monthly active users around the world, mostly in Greater China, Southeast Asia, North and South America. In 2016, XD launched TapTap, an application store for mobile games and gamers' community that now has 44 million monthly active users worldwide.

Industries: Gaming

- Reduces cross-region connection disruption by more than 90% with Premium Tier Network
- Lowers in-game latency through a single global IP address supported by Premium Tier Network
- Shortens game loading time with Cloud CDN
- Helps identify causes of in-game errors with data

Zero downtime during deployment

Location: China

(<https://cloud.google.com/compute/>), Network

Google Cloud (<https://cloud.google.com/>)
(<https://cloud.google.com/network-tiers/>),

BigQuery (<https://cloud.google.com/bigquery/>)
(<https://cloud.google.com/load-balancing/>),

Cloud CDN (<https://cloud.google.com/cdn/>)

Compute Engine
(<https://cloud.google.com/compute/>)

Network Service Tiers ALPHA
(<https://cloud.google.com/network-tiers/>)

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Cloud Load Balancing
(<https://cloud.google.com/load-balancing/>)
[Contact us](#) ([/contact](#)),

Virtual Private Cloud
(<https://cloud.google.com/vpc/>)

Cloud Storage (<https://cloud.google.com/storage/>)

**analytics
results
generated
by
BigQuery**

Online video games have become an important part of daily entertainment for many people. While there is an endless selection of games on the market that are easy to access, some consumers have noticed a growing number of games that appear to be too profit-oriented. These games either limit gamers from playing well without buying advanced features, or are overwhelmed with ads, which greatly undermines gaming experience.

XD (<https://www.xd.com/?lang=en>) has been on a mission to build high-quality games with passion and deliver the pure pleasure of gaming that can be enjoyed by all age groups. Founded in 2009, the global game developer and distributor has released dozens of web and mobile games, and currently offers nearly 40 free and paid games to 50 million monthly active users around the world.

"We focus primarily on game quality, instead of purely on revenues. This is why we only hire people who have a great passion for games. We want our users to be fully immersed in our well-designed games without being constantly bothered by overwhelming ads and payment requests," says Jonathan Chen, head of public relations at XD.

For example, Ragnarok M: Eternal Love

(<https://play.google.com/store/apps/details?id=com.gravity.romg&hl=en&gl=US>)

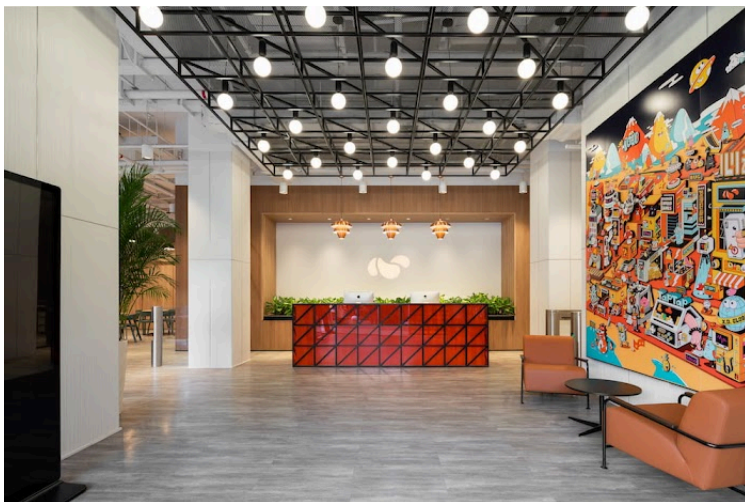
, one of XD's flagship games that was released in 2015, is widely praised by users for its captivating storyline and immersive sound effects. Unlike many massively multiplayer online role-playing games of its kind, the game encourages players to collaborate with each other, and provides ways for gamers to gain achievements while exploring its world without battling or purchasing advanced weapons.

As part of its commitment to bringing real quality games to gamers worldwide, XD launched TapTap (<https://www.taptap.com/mobile?hl=en>) in 2016, an app store that recommends mobile games based more on their quality gameplay rather than popularity. As of fiscal year 2021, the platform has 44 million monthly active users globally and continues to bring high-quality, exclusive games to gamers around the world. For example, TapTap provides the download and community service for Fortnite (<https://www.taptap.io/app/82354>), a top popular game worldwide, and partners with game developer Epic Games.

When XD released Ragnarok M: Eternal Love in Taiwan region, it deployed the game in a local on-prem environment, because all Taiwanese users' data are required to be stored locally according to the island's regulations. The game received great popularity, but the service was constantly down due to the limited network bandwidth provided by the local IT infrastructure vendor. To solve this issue, XD in 2017 decided to move Ragnarok M: Eternal Love

in Taiwan to Google Cloud (<https://cloud.google.com/>) for its high-performance network and data center in Taiwan.

"We wanted our users to be able to fully enjoy the fascinating adventures in our game by moving it to a stable and reliable cloud infrastructure," recalls Yufan Wang, head of IT operations at XD. "Google Cloud offers the best solution with its highly available global network and local data center."



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Achieving highly stable connectivity with VPC

XD now has its game servers for Ragnarok M: Eternal Love in Taiwan hosted on [Compute Engine](https://cloud.google.com/compute) (<https://cloud.google.com/compute>), and uses [Cloud Load Balancing](https://cloud.google.com/load-balancing) (<https://cloud.google.com/load-balancing>) to distribute traffic to ensure that its servers are not overloaded. Since all the network connections to and from its game servers go through a virtual private network supported by [Virtual Private Cloud](https://cloud.google.com/vpc) (<https://cloud.google.com/vpc>) (VPC) and [Premium Tier Network](https://cloud.google.com/network-tiers) (<https://cloud.google.com/network-tiers>), which guarantees more stable connectivity compared to the public internet and sufficient bandwidth at all times, XD has been able to provide

smoother gaming experiences. Previously, players experienced three to five downtimes every week because the network traffic exceeded the available bandwidth. After migrating to Google Cloud, no downtime has ever occurred.

With such excellent results, XD adopted the same cloud infrastructure for Flash Party

(https://www.taptap.io/app/211926?hreflang=en_US), its latest mobile game released in early 2022, in Japan and South America. Since Flash Party is a player-versus-player combat game, the company has to deploy its game servers in the same region where gamers are to provide faster loading speed, but all the servers still need to be connected with the database hosted in other regions like Singapore.

When XD first deployed Flash Party in South America in another public cloud, the cross-region connection went through the public internet, which led to network disruptions twice or three times per day, and occasional data loss. With the single private network of Google Cloud that spans all regions, connection disruption has been reduced to once or twice a week.

"VPC and Premium Tier Network have enabled us to greatly enhance the network stability of our games. Its high-performance cross-region private network is particularly helpful, because we can deploy our game servers anywhere to comply with local regulations or provide faster loading speed, without compromising the stability of data transferring across different regions," explains Wang.

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Reducing game loading time with Cloud CDN

Besides VPC and Premium Tier Network, another Google Cloud product that has greatly contributed to improving the network quality of XD's games is Cloud CDN (<https://cloud.google.com/cdn>), which the company leverages to deliver the game content stored in Cloud Storage (<https://cloud.google.com/storage>). Wang says that since Cloud CDN can cache content in servers that are physically closer to end users, the content download speed supported by Cloud CDN through wireless internet is as fast as the one through wired internet.

Additionally, Cloud CDN offers Anycast IP, which is an IP network addressing scheme that makes it possible for multiple servers to share a single global IP address. This way, network performance can be more consistent across different locations with lower latency. In 2021, XD moved Ragnarok M: Eternal Love in North America from another cloud platform to Google Cloud, because the game servers on the former cloud platform had different IP addresses, which delivered different levels of latency up to 150 microseconds. After the migration, 80% of

the gamers enjoy latency of less than 75 microseconds.

"Cloud CDN is crucial to our business, because it supports fast content delivery speeds, even when we need to send out an enormous amount of files," notes Wang. "The Anycast IP that it provides has also helped us further enhance the gaming experiences that we offer by minimizing in-game latency."



Facilitating troubleshooting and better services to users worldwide

For Ragnarok M: Eternal Love in Taiwan and North America, XD also uses BigQuery.

(<https://cloud.google.com/bigquery>) to analyze the data of user behavior in the game, such as how gamers earn in-game assets and how in-game transactions take place, with Cloud Storage to save archived data for lower data storage costs. Based on the analytics results generated by BigQuery, the XD team is able

to swiftly identify the cause when an error occurs and clear the issue.

"The data analytics supported by BigQuery has greatly benefited our game DevOps engineers, because they can now trace how an error happens in a more systematic and efficient way," says Wang.

After having successfully improved the gaming experiences of its two games by moving them to Google Cloud, XDa plans to use the infrastructure of Google Cloud to deploy its new games in regions outside China for better network performance. The company will also migrate TapTap in North America to the local data centers of Google Cloud to enhance content delivery speed for local users, and comply with local data requirements.

"The results that we have achieved with the high-performance global network of Google Cloud are truly amazing. By leveraging its powerful infrastructure, we'll be able to provide more high-quality games with minimum glitches to users around the world," concludes Wang.

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